

Point sets whose complete geometric graph needs $\frac{3}{4}n - O(\sqrt{n})$
crossing-free paths:
an affirmative answer to Problem 9 of Dumitrescu, Pach,
Saghafian and Scott

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Machine-generated note

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Abstract

Let A be a set of n points in the plane in general position and let $\pi(A)$ be the least number of crossing-free paths of the complete geometric graph $K_n[A]$ whose union contains every edge. A path has at most $n - 1$ edges, so $\pi(A) \geq \lceil n/2 \rceil$, and this is attained in convex position. Dumitrescu, Pach, Saghafian and Scott asked (Problem 9 of [1]) whether some constant $c > 1/2$ satisfies $\pi(A) \geq cn$ for infinitely many n . We answer this affirmatively. The vertices of a regular $(6q - 1)$ -gon together with its centre form a $6q$ -point set with $\pi(A) \geq 4q = \frac{2}{3}n$, and an odd regular h -gon with a cluster of r points near its centre satisfies $\pi(A) \geq \lceil (3r - 2)h/(4r - 2) \rceil$; taking $r \approx \sqrt{n}$ gives n -point sets with $\pi(A) \geq \frac{3}{4}n - \sqrt{n} - 1$ for every $n \geq 9$, so every $c < 3/4$ works. The engine is a simultaneous incompatibility: the h long chords of the polygon pairwise cross unless they share a vertex, so a plane path contains at most two of them, and then only as a wedge that encloses the cluster and blocks every spoke from the cluster to any other polygon vertex. No matching upper bound is given, so the constant $3/4$ is not shown to be optimal.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

A finite point set $A \subset \mathbb{R}^2$ is in *general position* if no three of its points are collinear. The *complete geometric graph* $K_n[A]$ on an n -point set A has vertex set A and all $\binom{n}{2}$ straight segments between points of A as edges. Two edges *cross* if they share a point that is interior to both; a set of edges is *plane* or *crossing-free* if no two of its edges cross. A *path* is an ordinary simple graph-theoretic path $x_0x_1 \cdots x_\ell$ on distinct points of A , drawn with straight edges; it need not visit every point of A . We write

$$\pi(A) = \min\{k : \text{there are crossing-free paths } P_1, \dots, P_k \text{ of } K_n[A] \\ \text{with } E(P_1) \cup \dots \cup E(P_k) = E(K_n[A])\}.$$

Covering paths may share edges and may be short. This is the most permissive reading of “covering the edge set by crossing-free paths”, and therefore the reading under which a lower bound is hardest to prove: a lower bound on $\pi(A)$ implies the same bound for partitions into plane paths and for covers by plane spanning (Hamiltonian) paths.

Since a path on at most n vertices has at most $n - 1$ edges, every point set satisfies the trivial bound

$$\pi(A) \geq \left\lceil \binom{n}{2} / (n - 1) \right\rceil = \left\lceil \frac{n}{2} \right\rceil, \quad (1)$$

and for points in convex position this is an equality: the classical zig-zag Hamiltonian paths, used by Bernhart and Kainen [2] to compute the book thickness of K_n , partition the edges of a convex $2m$ -gon into m plane paths (the referee re-verified $\pi = \lceil n/2 \rceil$ for convex $n = 4, \dots, 8$ by exhaustive search, Appendix B).

Dumitrescu, Pach, Saghaian and Scott [1] study coverings of complete geometric graphs by crossing-free paths and matchings: the edge set of every complete geometric graph can be partitioned into $O(n^{3/2})$ crossing-free paths (or matchings), a random n -point set admits a cover by $O(n \log n)$ crossing-free paths with high probability, while some point sets need a quadratic number of *monotone* paths. In their concluding remarks (Section 5, remark 2) they ask:

Question 1.1 ([1, Problem 9]). Does there exist a constant $c > 1/2$ with the property that, for infinitely many values of n , there are n -element point sets A in the plane such that every covering of the edge set of $K_n[A]$ by crossing-free paths requires at least cn paths?

(The problem catalog from which this target was drawn lists the question as “Problem 5.1”; that label is a section-based renumbering artefact, and the statement above is verbatim from the source.) In [1] Question 1.1 is presented as a first step towards their Problem 1, which asks for linear *upper* bounds; an affirmative answer to Question 1.1 says nothing about that problem.

Related work. The closest well-studied quantity is the minimum number $\chi_{\text{cross}}(A)$ of plane subgraphs into which $E(K_n[A])$ can be partitioned. A cover by k plane paths yields a partition into at most k plane subgraphs (assign each edge to one path containing it), so

$$\chi_{\text{cross}}(A) \leq \pi(A) \quad \text{for every } A, \quad (2)$$

and a large value of π carries no information about χ_{cross} . For χ_{cross} the trivial bound is again $\lceil n/2 \rceil$; the best lower bound in the literature is the additive $n/2 + 1$ of Obenaus and Orthaber [3] for their “bumpy wheel” point sets, continued in [4] and complemented by upper bounds for dense point sets by Dumitrescu and Pach [6]. Pach, Saghaian and Schnider [5] prove that decomposing the complete *convex* geometric graph into plane *star-forests* requires $n - 1$ of them. (The catalog entry for this problem attributes to [5] an upper bound of $n/2 + o(n)$ plane subgraphs for convex point sets; that is a misreading of the source: the figure $n/2 + o(n)$ is the speculation of the authors of [1] about plane subgraphs of general point sets, and [5] proves the $n - 1$ lower bound for star-forests just stated.) The paper [7] on “plane covering paths” concerns covering the *points* of a set by the vertices of few plane paths, a different problem. The referee found no linear lower bound with a constant exceeding $1/2$ for either π or χ_{cross} in any independent source, so the point sets below appear to be the first with $\pi(A) \geq cn$ for some $c > 1/2$. Consistently with (2), they do *not* beat $n/2$ for χ_{cross} (Remark 4.6).

2 The construction and the main result

Fix an odd integer $h = 2m + 1 \geq 5$ and let

$$v_j = \left(\cos \frac{2\pi j}{h}, \sin \frac{2\pi j}{h} \right), \quad 0 \leq j < h,$$

be the vertices of the regular h -gon inscribed in the unit circle, with indices read modulo h , and $V = \{v_0, \dots, v_{h-1}\}$. The *long chords* are

$$D = \{v_i v_{i+m} : 0 \leq i < h\},$$

a set of h distinct segments (for $h = 5$, the pentagram). Because $-m \equiv m + 1 \pmod{h}$, the two long chords at v_i are $v_i v_{i+m}$ and $v_i v_{i+m+1}$, and their endpoints v_{i+m}, v_{i+m+1} are adjacent on the polygon. Put

$$T_i = \text{conv}\{v_i, v_{i+m}, v_{i+m+1}\}, \quad U = \bigcap_{i=0}^{h-1} \text{int}(T_i).$$

For a finite set $B \subset U$ (the *cluster*) put $A = V \cup B$ and call the edges

$$S = \{vx : v \in V, x \in B\}$$

the *spokes*. For a path P of $K_n[A]$ write $d(P) = |E(P) \cap D|$ and $s(P) = |E(P) \cap S|$.

Theorem 2.1 (Main theorem). *Let $h \geq 5$ be odd and let V, D, U, S be as above.*

(1) *The set $A = V \cup \{o\}$, $o = (0, 0)$, is in general position, has $n = h + 1$ points, and satisfies*

$$\pi(A) \geq \left\lceil \frac{2h}{3} \right\rceil.$$

In particular, for $h = 6q - 1$ the set A has $n = 6q$ points and $\pi(A) \geq 4q = \frac{2}{3}n$, for every $q \geq 1$.

(2) *Let $r \geq 2$. There is an r -point set $B \subset U$ such that $A = V \cup B$ is in general position, and every such A (with $n = h + r$ points) satisfies*

$$\pi(A) \geq \left\lceil \frac{(3r-2)h}{4r-2} \right\rceil.$$

(3) *Consequently, for every $n \geq 9$ there is an n -point set A in general position with*

$$\pi(A) \geq \frac{3}{4}n - \sqrt{n} - 1,$$

so in particular $\pi(A) \geq \frac{3}{4}n - O(\sqrt{n})$.

Corollary 2.2. *The answer to [Question 1.1](#) is affirmative. The constant $c = 2/3$ works for every $n = 6q$, $q \geq 1$, and every constant $c < 3/4$ works for all sufficiently large n . The trivial bound (1) is therefore not tight up to a constant factor for general point sets.*

Proof. Part (1) of [Theorem 2.1](#) gives $6q$ -point sets with $\pi(A) \geq \frac{2}{3} \cdot 6q$ for every $q \geq 1$, which is the assertion for $c = 2/3$ and infinitely many n . For $c < 3/4$, part (3) gives $\pi(A) \geq \frac{3}{4}n - \sqrt{n} - 1 \geq cn$ as soon as $(\frac{3}{4} - c)n \geq \sqrt{n} + 1$, which holds for all large n . $\square$

Idea of proof. The long chords of an odd polygon pairwise cross unless they share a vertex ([Lemma 3.1](#)), so a plane path contains at most two of them, and if it contains two they form a wedge $v_i v_{i+m}, v_i v_{i+m+1}$ at a common vertex. The wedge together with the polygon side $v_{i+m} v_{i+m+1}$ bounds the triangle T_i , which contains the whole cluster $B \subset U$; hence a segment from a cluster point to any polygon vertex other than the three vertices of the wedge must leave T_i through one of the two chords and crosses it ([Lemma 3.2](#)). So a plane path may be rich in chords or rich in spokes, but never both: with two chords it has at most two spokes, since v_i is already saturated and v_{i+m}, v_{i+m+1} can each take one more edge ([Lemma 3.4](#)). For $r = 1$ this already gives $|E(P) \cap (D \cup S)| \leq 3$ for every plane path P , while $|D \cup S| = 2h$, whence $\pi \geq \lceil 2h/3 \rceil$. For general r one weights chords by $2r - 2$ and spokes by 1: every plane path has weight at most $4r - 2$, whereas the total weight is $(3r - 2)h$, and the standard fractional-cover bound ([Lemma 3.3](#)) gives $\pi \geq \lceil (3r - 2)h / (4r - 2) \rceil$. Since $(3r - 2)/(4r - 2) = \frac{3}{4} - \frac{1}{8r - 4}$, choosing $r \approx \sqrt{n}$ with $h = n - r$ odd balances the loss $\frac{3}{4}r$ from the cluster against the loss $(n - r)/(8r - 4)$ from the weighting and yields $\frac{3}{4}n - O(\sqrt{n})$. [\[full proof\]](#)

3 The four lemmas

Lemma 3.1 (Long chords). *Let $h = 2m + 1 \geq 5$. Two vertex-disjoint edges of D cross. Consequently every crossing-free path P of $K_n[A]$ (for any $A \supseteq V$) satisfies $d(P) \leq 2$, and if $d(P) = 2$ then the two chords of P in D are $v_i v_{i+m}$ and $v_i v_{i+m+1}$ for some i .*

Idea of proof. By rotational symmetry it suffices to look at v_0v_m . A chord v_iv_{i+m} vertex-disjoint from it has $i \notin \{0, m, m+1\}$, and a check of the indices shows that one of its endpoints lies in $\{v_1, \dots, v_{m-1}\}$ and the other in $\{v_{m+1}, \dots, v_{2m}\}$; the endpoints alternate with v_0, v_m around the strictly convex polygon, so the chords cross. Among any three edges of a simple path two are vertex-disjoint (the path has maximum degree 2 and contains no triangle), so a plane path holds at most two chords, and two chords in it must share a vertex. [\[full proof\]](#)

Lemma 3.2 (Wedge). *Let $h = 2m + 1 \geq 5$. The origin lies in the interior of every T_i , so U is an open neighbourhood of the origin. Let $x \in U$ be such that $V \cup \{x\}$ has no three collinear points, let $0 \leq i < h$, and put $a = v_{i+m}, b = v_{i+m+1}$. Then for every $z \in V \setminus \{v_i, a, b\}$ the segment xz crosses v_ia or v_ib .*

Idea of proof. Rotating v_i to $(1, 0)$ puts a, b at $(-\cos \frac{\pi}{h}, \pm \sin \frac{\pi}{h})$, and the origin is visibly inside that triangle. The point z is a vertex of the convex polygon other than v_i, a, b , hence outside T_i , while x is inside; so xz leaves T_i through one of its sides. It cannot leave through ab , a side of the outer polygon with x and z both strictly on its inner side. So it meets v_ia or v_ib , and general position makes this a proper crossing. [\[full proof\]](#)

Lemma 3.3 (Weighted counting). *Let $w: E(K_n[A]) \rightarrow \mathbb{R}_{\geq 0}$, let $W = \sum_e w(e)$, and suppose every crossing-free path P satisfies $\sum_{e \in E(P)} w(e) \leq L$ for some $L > 0$. Then $\pi(A) \geq \lceil W/L \rceil$.*

Idea of proof. If $P_1, \dots, P_k$ cover every edge, then $W \leq \sum_j w(P_j) \leq kL$. This is the dual certificate of the fractional covering linear program; it is standard and was already recorded, as Proposition 4.1, in the first attempt on this problem described in [Section 5](#). [\[full proof\]](#)

Lemma 3.4 (Path budget). *Let $h \geq 5$ be odd, $r \geq 1$, and let $B \subset U$ with $|B| = r$ be such that $A = V \cup B$ is in general position. Every crossing-free path P of $K_n[A]$ satisfies*

- (i) $d(P) \leq 2$;
- (ii) $s(P) \leq 2r$;
- (iii) if $d(P) = 2$ then $s(P) \leq 2$;
- (iv) if $r = 1$ then $|E(P) \cap (D \cup S)| \leq 3$.

Idea of proof. (i) is [Lemma 3.1](#); (ii) holds because every spoke has exactly one endpoint in B and P has degree at most 2 there. For (iii), the two chords are v_ia, v_ib by [Lemma 3.1](#), and by [Lemma 3.2](#) every spoke of P ends at v_i, a or b ; v_i already has degree 2 in P , and a, b have degree 1, so at most one spoke each. For (iv) with $B = \{o\}$: if $d(P) \leq 1$ then $s(P) = \deg_P(o) \leq 2$; if $d(P) = 2$ then the spokes oa, ob cannot both lie in P , since o, a, v_i, b, o would be a cycle. [\[full proof\]](#)

4 Remarks

Remark 4.1 (Interpretation). The referee checked the statement of [Question 1.1](#) verbatim against arXiv:2507.10840v2 (Problem 9, Section 5, p. 8; also Comb. Number Th. 15 (2026) 73–82). The writeup uses the weakest available notion of π : covering paths may overlap, need not be spanning, and are ordinary simple paths drawn plane. Allowing overlaps makes π smaller than the partition version and allowing short paths makes it smaller than a spanning-path version, so [Theorem 2.1](#) implies the corresponding bounds for every stricter version. The constructed sets are in general position, the standing hypothesis of [\[1\]](#), and this was verified in exact arithmetic ([Remark 4.5](#)).

Remark 4.2 (Lower bound only). No upper bound on $\pi(A)$ for these point sets, or on $\max_A \pi(A)$, is given, so the constant $3/4$ is not shown to be optimal. It is an artefact of the three constraints $d \leq 2$, $s \leq 2r$, $d = 2 \Rightarrow s \leq 2$ and of the weighting that balances them; the true value of $\max_{|A|=n} \pi(A)$ remains open between $\frac{3}{4}n - O(\sqrt{n})$ and the $O(n^{3/2})$ of [\[1\]](#).

Remark 4.3 (Quantifying “sufficiently large”). The original writeup states part (3) as $\pi(A) \geq \frac{3}{4}n - O(\sqrt{n})$ for all sufficiently large n . The explicit form $\frac{3}{4}n - \sqrt{n} - 1$ for $n \geq 9$ in [Theorem 2.1](#) comes from the concrete choice $r \in \{\lfloor \sqrt{n} \rfloor, \lfloor \sqrt{n} \rfloor + 1\}$ making $h = n - r$ odd ([Appendix A](#)); it is not optimised. Optimising r (the optimum is $r \approx \sqrt{n/6}$) the referee obtained $\pi(A) \geq \frac{3}{4}n - 0.61\sqrt{n}$ throughout $n = 10^2, \dots, 10^6$, and the thresholds $c = 0.6$ from $n \geq 13$, $c = 0.7$ from $n \geq 120$, $c = 0.74$ from $n \geq 3600$, $c = 0.749$ from $n \geq 373\,753$. None of this is needed for [Question 1.1](#), which part (1) answers with $c = 2/3$ for every $n = 6q$, $q \geq 1$, starting at $n = 6$.

Remark 4.4 (The two parts are separate arguments). The bound of part (2) is valid for $r = 1$ as well, but then the chord weight $2r - 2$ vanishes and it degenerates to $\lceil h/2 \rceil$, weaker than the $\lceil 2h/3 \rceil$ of part (1). Part (1) uses the extra fact [Lemma 3.4\(iv\)](#), that with two chords a single centre admits only *one* spoke (two would close a 4-cycle), and is not a special case of part (2). Oddness of h is load-bearing in two places: it makes the two long chords at each vertex distinct, and it keeps the centre off every line through two polygon vertices (for even h the centre is collinear with each antipodal pair; the referee’s harness confirms $h = 6$ is degenerate). The region U shrinks with h (it contains the disc of radius $\sin \frac{\pi}{2h}$ about the origin, the distance from the centre to a long chord), but no quantitative bound is needed: for each fixed h it is a fixed non-empty open set.

Remark 4.5 (Computational checks by the referee). The referee’s scripts are in the repository `verification_astra/scripts/2507.10840__01/`. All geometric predicates were verified in exact arithmetic twice: once on the true regular polygon with sympy algebraic numbers, and once, in an independent implementation, on rational near-regular coordinates (denominator 10^7 ; cluster points rational, in the disc of radius $\frac{1}{2} \sin \frac{\pi}{2h}$) where every test is an integer determinant. Since the properties in [Lemmas 3.1](#) and [3.2](#) are open conditions, the rational configurations that pass are themselves witnesses, so [Theorem 2.1](#) also has witnesses with rational coordinates. The checks (no three collinear, convexity, both directions of [Lemma 3.1](#), every cluster point strictly inside every T_i , and the crossing conclusion of [Lemma 3.2](#) for all triples x, i, z) pass for $(h, r) = (5, 1), (5, 2), (7, 1), (7, 2), (7, 3), (9, 1), (9, 2), (9, 3), (11, 1), (13, 1), (15, 3), (21, 4), (25, 5)$. Exhaustive enumeration of all crossing-free paths for $(h, r) = (5, 1), (7, 1), (9, 1), (11, 1), (5, 2), (7, 2), (9, 2), (7, 3)$ found $\max d(P) = 2$, $\max s(P) = 2r$ (attained for $d \in \{0, 1\}$), $\max\{s(P) : d(P) = 2\} = 2$ for $r \geq 2$ and $= 1$ for $r = 1$, maximum weight exactly $4r - 2$, and $\max |E(P) \cap (D \cup S)| = 3$ for $r = 1$: every inequality of [Lemma 3.4](#) holds with equality. Finally, exact integer set cover over all maximal crossing-free paths (130, 1449 and 15 723 of them) gives

$$\pi(A) = 4, 5, 6 \quad \text{for the 5-, 7-, 9-gon with centre } (n = 6, 8, 10),$$

against the trivial bounds 3, 4, 5 and matching $\lceil 2h/3 \rceil$ exactly; for the 5-gon with two cluster points ($n = 7$, 633 maximal paths) $\pi = 4 = \lceil n/2 \rceil$. The values for $n = 6, 8$ were obtained by two independent solvers (scipy milp on the exact geometry and a branch-and-bound set-cover solver on the rational geometry); the value for $n = 10$ rests on scipy milp alone, because the branch-and-bound cross-check hit its wall-clock cap. The branch-and-bound solver returns $\pi = 2, 3, 3, 4, 4$ for convex $n = 4, \dots, 8$, the known value $\lceil n/2 \rceil$.

Remark 4.6 (No conflict with the plane-subgraph record). By (2), π dominates the plane-subgraph partition number, and the gain here is specific to paths: the referee computed exactly $\chi_{\text{cross}} = 3 = n/2$ for the 5-gon with centre and $\chi_{\text{cross}} = 4 = n/2$ for the 7-gon with centre, while $\pi = 4$ and 5 respectively. These point sets therefore do not improve on the $n/2 + 1$ of Obenaus and Orthaber [3] for plane subgraphs, and the result is not in tension with that record.

Remark 4.7 (Relation to the first attempt). The first attempt on this problem (by gpt-5.6-so1, supplied as context; see [Section 5](#)) proved no lower bound above $n/2$ but isolated why elementary arguments stop there: nonnegative combinations of vertex-degree constraints and of “at most one edge from a pairwise-crossing matching” constraints, each bounded separately, never exceed $n/2$. The present argument is consistent with that obstruction and explains how it is bypassed: D is not a matching (every polygon vertex lies on two long chords), and [Lemma 3.4\(iii\)](#) is precisely the simultaneous incompatibility between two constraint families that the first attempt identified as the missing ingredient.

Remark 4.8 (Novelty). Novelty was checked only against what could be found online on 2026-09-17. arXiv:2507.10840 had no recorded citations in Semantic Scholar or OpenAlex, and the referee ruled out [3, 4, 6, 5, 7] and arXiv:2508.17277, arXiv:2510.22365, arXiv:2402.11044 as containing the construction or the bound. The single web hit for the construction is a pull request opened by this project on 2026-09-16, which is this campaign’s own output and not prior art. A human check of the literature is still required.

5 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model gpt-6-astra (reasoning effort max) in a single autonomous pass started on 2026-09-17, in the attacks_retry leg of a sweep over open problems catalogued from arXiv (catalog id 2507.10840__01; writeup in attacks_retry/2507.10840__01/output.md). This was a second attempt: the prompt contained an earlier attempt by gpt-5.6-sol (verdict “partial”, no construction), which proved the convex-position value $\lceil n/2 \rceil$, the weighted-counting bound of [Lemma 3.3](#) (its Proposition 4.1), and the obstruction of [Remark 4.7](#). The new writeup shares no argument with it beyond the standard [Lemma 3.3](#), and nothing inherited from it is presented here as new. (2) An independent adversarial referee agent (claude-opus-5, 2026-09-17) checked the problem statement against the source, re-derived all twenty steps of the writeup, searched the literature for prior resolutions, and verified the geometry and the bounds computationally as described in [Remark 4.5](#); its verdict was CONFIRMED, high confidence, with no step marked as a gap. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1: the text was reorganised around [Theorem 2.1](#) and the four lemmas of [Section 3](#), with full proofs moved to [Appendix A](#). Every deviation from the writeup is listed here.

- The general-position argument for the cluster read “avoiding the finitely many lines determined by previously chosen points”; following the referee, it now reads “lines determined by V and the previously chosen points” ([Appendix A](#)).
- “For every sufficiently large n , $\pi(A) \geq \frac{3}{4}n - O(\sqrt{n})$ ” has been quantified as $\pi(A) \geq \frac{3}{4}n - \sqrt{n} - 1$ for every $n \geq 9$, via the explicit choice $r \in \{\lfloor \sqrt{n} \rfloor, \lfloor \sqrt{n} \rfloor + 1\}$ with $h = n - r$ odd ([Appendix A](#), [Remark 4.3](#)). The writeup’s own asymptotic statement is recovered as a special case.
- Two steps the writeup stated in a clause are proved in full: that chords of a strictly convex polygon with alternating endpoints cross ([Appendix A](#)), and that any three edges of a simple path contain two vertex-disjoint ones ([Appendix A](#)). The proper-crossing conclusion of [Lemma 3.2](#) is derived explicitly from general position ([Appendix A](#)).
- The weighted-counting step, done inline in the writeup, is stated as [Lemma 3.3](#) and attributed to standard fractional-cover duality and to the first attempt.
- Added, from the referee report: the statement that no upper bound is given ([Remark 4.2](#)); the interpretation discussion ([Remark 4.1](#)); the relation between the two parts and the role of oddness ([Remark 4.4](#)); the computational checks ([Remark 4.5](#)); the plane-subgraph comparison ([Remark 4.6](#)); the literature [2, 3, 4, 6, 5, 7] and the correction of the catalog’s numbering and of its attribution to [5] ([Question 1.1](#) and the related-work paragraph). The referee report refers to arXiv:2405.17172 as “Dumitrescu–Toth”; arXiv lists its authors as Dumitrescu and Pach, and it is cited here as [6].

No other mathematical content was changed. **No human mathematician has checked this argument.** We circulate it for human review, not as a claim to a new resolution.

A Full proofs

Throughout, $h = 2m + 1 \geq 5$, indices are modulo h , and V, D, T_i, U are as in [Section 2](#). Since the v_j lie on a circle, no three points of V are collinear, and V is in strictly convex position: for any two

vertices v_p, v_q , the vertices on the two open arcs between them lie strictly on the two sides of the line $v_p v_q$.

Lemma 3.1 (Long chords). *Let $h = 2m + 1 \geq 5$. Two vertex-disjoint edges of D cross. Consequently every crossing-free path P of $K_n[A]$ (for any $A \supseteq V$) satisfies $d(P) \leq 2$, and if $d(P) = 2$ then the two chords of P in D are $v_i v_{i+m}$ and $v_i v_{i+m+1}$ for some i .*

Proof. D has h elements. If $v_i v_{i+m} = v_j v_{j+m}$ as sets with $i \neq j$, then $i \equiv j + m$ and $j \equiv i + m$, so $2m \equiv 0 \pmod{h}$, impossible for $h = 2m + 1$. The rotation $v_j \mapsto v_{j+1}$ maps D onto itself.

Disjoint chords cross. By rotational symmetry it suffices to treat chords disjoint from $e_0 = v_0 v_m$. A chord $v_i v_{i+m}$ is vertex-disjoint from e_0 iff $i \notin \{0, m\}$ and $i + m \notin \{0, m\}$, i.e. iff $i \notin \{0, m, m + 1\}$ (using $-m \equiv m + 1$). If $1 \leq i \leq m - 1$ then $m + 1 \leq i + m \leq 2m - 1$. If $m + 2 \leq i \leq 2m$ then $i + m \equiv i - m - 1 \in \{1, \dots, m - 1\}$. In both cases one endpoint lies in $\{v_1, \dots, v_{m-1}\}$, the open arc from v_0 to v_m , and the other in $\{v_{m+1}, \dots, v_{2m}\}$, the open arc from v_m to v_0 . By strict convexity these two endpoints lie strictly on opposite sides of the line ℓ through v_0, v_m , so the chord meets ℓ in a single point p interior to the chord. The point p lies in $\text{conv } V \cap \ell$, which is the segment $v_0 v_m$ (a chord of a convex polygon), and $p \neq v_0, v_m$ because otherwise three points of V would be collinear. Hence p is interior to both segments, i.e. the chords cross.

Consequences for a plane path P . Let e, f, g be three distinct edges of P . If every two of them shared a vertex, then either all three contain a common vertex, which would have degree 3 in P , or the three pairwise intersections are distinct vertices and e, f, g form a triangle, a cycle in P ; both are impossible in a simple path. So two of e, f, g are vertex-disjoint. If P contained three chords of D , two of them would be vertex-disjoint and would cross, contradicting planarity; hence $d(P) \leq 2$. If P contains exactly two chords, they are not vertex-disjoint, so they share a vertex v_i and are the two chords of D at v_i , namely $v_i v_{i+m}$ and $v_i v_{i+m+1}$ (the chord $v_{i-m} v_i$ equals $v_{i+m+1} v_i$). $\square$

Lemma 3.2 (Wedge). *Let $h = 2m + 1 \geq 5$. The origin lies in the interior of every T_i , so U is an open neighbourhood of the origin. Let $x \in U$ be such that $V \cup \{x\}$ has no three collinear points, let $0 \leq i < h$, and put $a = v_{i+m}, b = v_{i+m+1}$. Then for every $z \in V \setminus \{v_i, a, b\}$ the segment xz crosses $v_i a$ or $v_i b$.*

Proof. The origin is interior to T_i . Rotate the plane about the origin so that v_i becomes $(1, 0)$. Since $a = v_{i+m}$ and $b = v_{i+m+1}$ are at angular distances $2\pi m/h = \pi - \pi/h$ and $2\pi(m+1)/h = \pi + \pi/h$ from v_i , they become $(-\cos \frac{\pi}{h}, \sin \frac{\pi}{h})$ and $(-\cos \frac{\pi}{h}, -\sin \frac{\pi}{h})$. Write $c = \cos \frac{\pi}{h} > 0$, $s = \sin \frac{\pi}{h} > 0$. The triangle with vertices $(1, 0), (-c, s), (-c, -s)$ is symmetric about the x -axis; the side ab is the vertical segment on the line $x = -c < 0$, and the side from $(1, 0)$ to $(-c, s)$ passes over $x = 0$ at height $s/(1+c) > 0$. Hence $(0, 0)$ lies strictly between the line $x = -c$ and the two slanted sides, i.e. in $\text{int}(T_i)$. As U is a finite intersection of open sets each containing the origin, U is an open neighbourhood of the origin.

The crossing. Let $x \in U$ with $V \cup \{x\}$ having no three collinear points, and let $z \in V \setminus \{v_i, a, b\}$. Every point of V is an extreme point of the convex polygon $\text{conv } V$; since $T_i \subseteq \text{conv } V$ and z is not one of the three vertices of T_i , $z \notin T_i$ (an extreme point of $\text{conv } V$ lying in the triangle T_i would be a convex combination of the three vertices of T_i , hence equal to one of them). Thus the segment xz joins the interior point x of T_i to the exterior point z , and therefore contains a point p of the boundary of T_i with $p \neq x, z$. The boundary of T_i is the union of the three sides $ab, v_i a, v_i b$. The side ab is a side of the polygon $\text{conv } V$, so its line is a supporting line of $\text{conv } V$ and every other point of $\text{conv } V$ lies strictly on one side of it; both x (an interior point of $\text{conv } V$) and z (a vertex different from a, b) do, hence the whole segment xz avoids the line ab . So p lies on $v_i a$ or on $v_i b$, say on $v_i a$. If p were an endpoint of $v_i a$, then x, z and that endpoint would be three collinear points of $V \cup \{x\}$, excluded; so p is interior to $v_i a$, and it is interior to xz because $p \neq x, z$. The two segments are not collinear (that would make x, v_i, a collinear), so p is their unique common point and xz crosses $v_i a$. $\square$

Lemma 3.3 (Weighted counting). *Let $w: E(K_n[A]) \rightarrow \mathbb{R}_{\geq 0}$, let $W = \sum_e w(e)$, and suppose every crossing-free path P satisfies $\sum_{e \in E(P)} w(e) \leq L$ for some $L > 0$. Then $\pi(A) \geq \lceil W/L \rceil$.*

Proof. Let $P_1, \dots, P_k$ be crossing-free paths with $E(P_1) \cup \dots \cup E(P_k) = E(K_n[A])$. Every edge e lies in at least one P_j and $w \geq 0$, so $W = \sum_e w(e) \leq \sum_{j=1}^k \sum_{e \in E(P_j)} w(e) \leq kL$. Hence $k \geq W/L$, and k being an integer, $k \geq \lceil W/L \rceil$. Taking $k = \pi(A)$ gives the claim. $\square$

Lemma 3.4 (Path budget). *Let $h \geq 5$ be odd, $r \geq 1$, and let $B \subset U$ with $|B| = r$ be such that $A = V \cup B$ is in general position. Every crossing-free path P of $K_n[A]$ satisfies*

- (i) $d(P) \leq 2$;
- (ii) $s(P) \leq 2r$;
- (iii) if $d(P) = 2$ then $s(P) \leq 2$;
- (iv) if $r = 1$ then $|E(P) \cap (D \cup S)| \leq 3$.

Proof. Let P be a crossing-free path of $K_n[A]$, $A = V \cup B$.

(i) is the second assertion of [Lemma 3.1](#).

(ii) Every spoke has exactly one endpoint in B , so $s(P) \leq \sum_{x \in B} \deg_P(x) \leq 2r$.

(iii) Suppose $d(P) = 2$. By [Lemma 3.1](#) the two chords are $v_i a$ and $v_i b$ with $a = v_{i+m}$, $b = v_{i+m+1}$. Let $xz \in E(P)$ be a spoke, $x \in B \subset U$, $z \in V$. As A is in general position, $V \cup \{x\}$ has no three collinear points, so if $z \notin \{v_i, a, b\}$ then by [Lemma 3.2](#) the edge xz crosses $v_i a$ or $v_i b$, both edges of P , contradicting planarity. Hence $z \in \{v_i, a, b\}$. The vertex v_i is incident with the two chords, so $\deg_P(v_i) = 2$ and no spoke of P ends at v_i . Each of a and b is incident with one chord of P and has degree at most 2, so at most one spoke of P ends at a and at most one at b . Therefore $s(P) \leq 2$.

(iv) Let $r = 1$, $B = \{o\}$. If $d(P) \leq 1$, then $|E(P) \cap (D \cup S)| \leq 1 + s(P) \leq 1 + \deg_P(o) \leq 3$. If $d(P) = 2$, with chords $v_i a, v_i b$ as above, the argument of (iii) shows that every spoke of P is oa or ob . Both cannot belong to P : together with $v_i a$ and $v_i b$ they would form the cycle o, a, v_i, b, o , and a path contains no cycle. So $s(P) \leq 1$ and $|E(P) \cap (D \cup S)| \leq 3$. $\square$

Theorem 2.1 (Main theorem). *Let $h \geq 5$ be odd and let V, D, U, S be as above.*

- (1) *The set $A = V \cup \{o\}$, $o = (0, 0)$, is in general position, has $n = h + 1$ points, and satisfies*

$$\pi(A) \geq \left\lceil \frac{2h}{3} \right\rceil.$$

In particular, for $h = 6q - 1$ the set A has $n = 6q$ points and $\pi(A) \geq 4q = \frac{2}{3}n$, for every $q \geq 1$.

- (2) *Let $r \geq 2$. There is an r -point set $B \subset U$ such that $A = V \cup B$ is in general position, and every such A (with $n = h + r$ points) satisfies*

$$\pi(A) \geq \left\lceil \frac{(3r-2)h}{4r-2} \right\rceil.$$

- (3) *Consequently, for every $n \geq 9$ there is an n -point set A in general position with*

$$\pi(A) \geq \frac{3}{4}n - \sqrt{n} - 1,$$

so in particular $\pi(A) \geq \frac{3}{4}n - O(\sqrt{n})$.

Proof. (1) *General position.* No three points of V are collinear. A line through o and a vertex v_j meets the unit circle exactly in v_j and $-v_j$, and $-v_j \in V$ would require $\pi \equiv 2\pi k/h \pmod{2\pi}$ for some integer k , i.e. $h = 2k \pmod{2h}$, impossible for odd h . So $A = V \cup \{o\}$ is in general position.

The bound. Every chord has both endpoints in V and every spoke has an endpoint in B , so $D \cap S = \emptyset$ and $|D \cup S| = h + h = 2h$. Let $P_1, \dots, P_k$ be crossing-free paths covering $E(K_n[A])$. Each P_j contains at most 3 edges of $D \cup S$ by Lemma 3.4(iv), and every edge of $D \cup S$ lies in some P_j , so $3k \geq 2h$, i.e. $\pi(A) \geq \lceil 2h/3 \rceil$. (Equivalently, apply Lemma 3.3 with the indicator weight of $D \cup S$ and $L = 3$.) For $h = 6q - 1$ we get $n = h + 1 = 6q$ and $\lceil (12q - 2)/3 \rceil = \lceil 4q - \frac{2}{3} \rceil = 4q = \frac{2}{3}n$; here $h \geq 5$ for every $q \geq 1$.

(2) *Existence of B .* By Lemma 3.2, U is a non-empty open subset of the plane. Choose $x_1, \dots, x_r$ successively: having chosen $x_1, \dots, x_{j-1}$, pick $x_j \in U$ off the finitely many lines determined by pairs of points of $V \cup \{x_1, \dots, x_{j-1}\}$ (the referee noted that the writeup's phrase "determined by previously chosen points" must include V ; a non-empty open set minus finitely many lines is non-empty). Then x_j is distinct from all earlier points (each lies on such a line) and is collinear with no two points of $V \cup \{x_1, \dots, x_{j-1}\}$. Hence $B = \{x_1, \dots, x_r\} \subset U$ has r points and $A = V \cup B$ is in general position.

The bound. Let $B \subset U$ be any r -point set with $A = V \cup B$ in general position, and define $w(e) = 2r - 2$ for $e \in D$, $w(e) = 1$ for $e \in S$, and $w(e) = 0$ for all other edges of $K_n[A]$ (all edges remain available to the covering paths). Since $D \cap S = \emptyset$, $|D| = h$ and $|S| = hr$,

$$W = (2r - 2)h + hr = (3r - 2)h.$$

For a crossing-free path P , $w(P) = (2r - 2)d(P) + s(P)$, and by Lemma 3.4(i)–(iii)

$$w(P) \leq \begin{cases} 2r, & d(P) = 0, \\ (2r - 2) + 2r = 4r - 2, & d(P) = 1, \\ 2(2r - 2) + 2 = 4r - 2, & d(P) = 2. \end{cases}$$

As $2r \leq 4r - 2$ for $r \geq 1$, every crossing-free path has $w(P) \leq L := 4r - 2 > 0$. Lemma 3.3 gives $\pi(A) \geq \lceil (3r - 2)h/(4r - 2) \rceil$.

(3) Let $n \geq 9$, put $k = \lfloor \sqrt{n} \rfloor \geq 3$, and let $r = k$ if $n - k$ is odd and $r = k + 1$ otherwise; set $h = n - r$. Then h is odd, $r \geq 3 \geq 2$, and

$$\sqrt{n} - 1 < k \leq r \leq k + 1 \leq \sqrt{n} + 1, \quad h = n - r \geq n - \sqrt{n} - 1 \geq 5$$

(the function $n - \sqrt{n} - 1$ is increasing and equals 5 at $n = 9$). By (2) there is an n -point set $A = V \cup B$ in general position with $\pi(A) \geq (3r - 2)h/(4r - 2)$. The identity

$$\frac{3r - 2}{4r - 2} = \frac{3}{4} - \frac{1}{8r - 4}$$

(clear denominators: $4(8r - 4) \cdot \frac{3r - 2}{4r - 2} = 8(3r - 2) = 3(8r - 4) - 4$) gives

$$\pi(A) \geq \left(\frac{3}{4} - \frac{1}{8r - 4} \right) (n - r) = \frac{3}{4}n - \frac{3}{4}r - \frac{n - r}{8r - 4}.$$

Now $\frac{3}{4}r \leq \frac{3}{4}(\sqrt{n} + 1)$, and since $8r - 4 > 8\sqrt{n} - 12 \geq 4\sqrt{n}$ for $n \geq 9$, also $\frac{n - r}{8r - 4} < \frac{n}{4\sqrt{n}} = \frac{\sqrt{n}}{4}$. Therefore

$$\pi(A) > \frac{3}{4}n - \frac{3}{4}\sqrt{n} - \frac{3}{4} - \frac{\sqrt{n}}{4} = \frac{3}{4}n - \sqrt{n} - \frac{3}{4} \geq \frac{3}{4}n - \sqrt{n} - 1.$$

In particular $\pi(A) \geq \frac{3}{4}n - O(\sqrt{n})$. □

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step numbers, equation numbers and section numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/2507.10840__01/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The problem. The prompt’s statement was checked verbatim against the source. In arXiv:2507.10840v2 (Dumitrescu, Pach, Saghaian, Scott; v1 14 Jul 2025, v2 10 Jan 2026; Comb. Number Th. 15 (2026) 73-82) the question is **Problem 9**, in Section 5 (“Concluding remarks”, remark 2, p. 8). The catalog’s label “Problem 5.1” is a section-based renumbering artifact, not a different problem. Verbatim:

Problem 9. Does there exist a constant $c > 1/2$ with the property that, for infinitely many values of n , there are n -element point sets A in the plane such that every covering of the edge set of $K_n[A]$ by crossing-free paths requires at least cn paths?

This is word-for-word the statement in `prompt.md`. The interpretation audit is therefore settled at the source level.

The writeup’s formalization. $\pi(A)$ = minimum number of crossing-free paths whose union contains every edge of $K_n[A]$; paths may overlap, need not be spanning, and are ordinary simple graph paths drawn as straight segments with no two of their own edges crossing. This is the *weakest* reading available, i.e. the one that makes a lower bound *hardest* to prove:

- allowing overlaps between covering paths makes π smaller than the partition version;
- allowing short (non-Hamiltonian) paths makes π smaller than a spanning-path version;
- “crossing-free” is taken as “plane as drawn”, the standard meaning.

So there is no interpretation gaming: the writeup proves a lower bound for the most permissive notion, which implies it for every stricter one. The constructed set is in general position (no three collinear), verified exactly (see below), so it also satisfies the paper’s standing hypothesis on A .

Fairness. A reasonable author of Problem 9 would consider it resolved affirmatively. The trivial bound is $\pi(A) \geq \lceil \text{binom}(n,2)/n \rceil = \lceil n/2 \rceil$, and it is tight for convex position (Bernhart-Kainen zig-zag; I re-verified $\pi = \lceil n/2 \rceil$ for convex $n = 4..8$ by exhaustive search, see Computational check). The writeup exhibits point sets with $\pi(A) \geq 2n/3$ for all $n = 6q$, and $\geq 3n/4 - O(\sqrt{n})$ for all large n . Any c in $(1/2, 3/4)$ works. That is exactly what Problem 9 asks for.

Relation to the previous (gpt-5.6-sol) attempt in the prompt. The new writeup shares nothing with it and inherits nothing unchecked. It is self-contained and re-derives, inline, the weighted-certificate duality that the earlier attempt called Proposition 4.1. It is also *consistent* with the earlier attempt’s obstruction (its Prop. 6.1: sums of degree constraints and pairwise-crossing-*matching* constraints cannot beat $n/2$): the family $\mathcal{D}$ used here is **not** a matching (every vertex has $\mathcal{D}$ -degree 2), and the new argument supplies precisely the “simultaneous incompatibility” that the earlier attempt identified as the missing ingredient ($d(P) = 2 \Rightarrow s(P) \leq 2$). No error from the earlier attempt is reproduced.

B.2 Step-by-step findings

Notation as in the writeup: $h = 2m+1 \geq 5$ odd, v_j the regular h -gon vertices, $\mathcal{D} = \{v_i v_{i+m}\}$ (h long chords), $\mathcal{T}_i = \text{conv}\{v_i, v_{i+m}\}$, $v_{i+m+1}\}$, $\mathcal{U} = \bigcap_i \text{int}(\mathcal{T}_i)$, $\mathcal{B} \subset \mathcal{U}$ the cluster of r points, $\mathcal{S} = \mathcal{V} \times \mathcal{B}$ the spokes, $d(P) = |\mathcal{E}(P) \cap \mathcal{D}|$, $s(P) = |\mathcal{E}(P) \cap \mathcal{S}|$.

step	label	note
1. $\mathcal{D}$ consists of h distinct chords	VALID	$v_i v_{i+m} = v_j v_{j+m}$ with $i \neq j$ forces $2m \equiv 0 \pmod{2m+1}$, false. Code asserts $ \mathcal{D} = h$. Re-derived independently. Edges of $\mathcal{D}$ disjoint from $v_0 v_m$ are those with $i \notin \{0, m, m+1\}$. For $i \in \{1..m-1\}$ the endpoints are $(i, i+m)$ with $i+m \in \{m+1..2m\}$; for $i \in \{m+2..2m\}$ the endpoints are $(i-m-1, i)$ with $i-m-1 \in \{1..m-1\}$. Either way one endpoint lies in the open arc (v_0, v_m) and the other in (v_m, v_0) , so the chords alternate and cross. Verified exactly for $h = 5, 7, 9, 11, 13, 15, 21, 25$ (and the converse: adjacent pairs do not cross).
2. Property 1: two vertex-disjoint edges of $\mathcal{D}$ cross	VALID	$-m \equiv m+1 \pmod{2m+1}$.
3. The two $\mathcal{D}$ -edges at v_i are $v_i v_{i+m}$ and $v_i v_{i+m+1}$	VALID	
4. Every crossing-free path has $d(P) \leq 2$, and if $d(P) = 2$ the two chords share a vertex	VALID	Among any three edges of a simple path two are vertex-disjoint (three pairwise-adjacent edges would form a triangle), and disjoint $\mathcal{D}$ -edges cross by step 2. Brute force: $\max d(P) = 2$ over all plane paths for every (h,r) tested.

step	label	note
5. Property 2, part a: the origin is strictly inside every T_i	VALID	Rotating v_i to $(1,0)$, the other two vertices are at angles $\pi \mp \pi/h$, i.e. $(-\cos(\pi/h), \pm \sin(\pi/h))$. The edge ab is the vertical line $x = -\cos(\pi/h) < 0$; on the edge from $(1,0)$ to $(-\cos(\pi/h), \sin(\pi/h))$ the height at $x = 0$ is $\sin(\pi/h)/(1+\cos(\pi/h)) > 0$, and symmetrically below. So 0 is strictly interior. Hence U is an open neighbourhood of the origin; it contains the disc of radius $\sin(\pi/(2h))$ (the distance from the centre to a long chord). Verified exactly.
6. Property 2, part b: for $x \in U, z \in V \setminus \{v_i, a, b\}$, segment xz properly crosses $v_i a$ or $v_i b$	VALID	$x \in \text{int } T_i$; z is a vertex of the strictly convex polygon not among the three, hence extreme in $\text{conv}(V) \supset T_i$ and so outside T_i . The segment must leave T_i ; it cannot cross side ab , since ab is an edge of the outer convex polygon and both x and z are strictly on its inner side. So it crosses $v_i a$ or $v_i b$, properly because general position excludes passing through v_i . Verified exhaustively (all x, i, z) for h up to 25, r up to 5.
7. General position of $V \cup \{o\}$ (Section 2)	VALID	Three points on a circle are never collinear; a line through the centre meets the circle in an antipodal pair, impossible for odd h . The harness's own self-test confirms the <i>even</i> case $h = 6$ is degenerate (v_0, o, v_3 collinear), so the check has teeth.
8. Inequality (1): $ E(P) \cap (D \cup S) \leq 3$ for $r = 1$	VALID	If $d(P) \leq 1$: $s(P) = \deg_P(o) \leq 2$, total ≤ 3 . If $d(P) = 2$ with chords $v_i a, v_i b$: by step 6 any spoke in P has outer endpoint in $\{v_i, a, b\}$; $o v_i$ is impossible ($\deg_P(v_i) = 2$ already); oa and ob cannot both occur (they would close the 4-cycle o, a, v_i, b, o inside a path). So $s(P) \leq 1$, total ≤ 3 . Brute force gives $\max = 3$ exactly, for $h = 5, 7, 9, 11$.
9. Bound (2): $\pi(A) \geq \text{ceil}(2h/3)$	VALID	D and S are disjoint, $ D \cup S = 2h$; each path covers ≤ 3 of them, so $3k \geq 2h$.
10. $h = 6q-1, n = 6q$ gives $\pi(A) \geq 4q = 2n/3$	VALID	$\text{ceil}((12q-2)/3) = 4q$; checked for $q = 1..2000$. Valid from $q = 1$ ($h = 5, n = 6$).
11. Existence of $B \subset U$ with $V \cup B$ in general position	VALID (wording nit)	U is open and 2-dimensional; pick the r points one at a time off the finitely many lines already determined. The writeup says "lines determined by previously chosen points", which should read "by V and the previously chosen points"; the intent is unambiguous and the conclusion is correct. Independently, I exhibited <i>explicit rational</i> point sets passing an exact no-three-collinear test for (h,r) up to $(25,5)$, so existence is not merely generic.
12. Inequality (3): $d(P) \leq 2, s(P) \leq 2r$	VALID	Second part: every S -edge has exactly one endpoint in B , so $s(P) = \sum_{x \in B} \deg_P(x) \leq 2r$. Attained in brute force for $d(P) \in \{0, 1\}$.
13. Inequality (4): $d(P) = 2 \Rightarrow s(P) \leq 2$	VALID	By step 6 all spokes of P end at v_i, a or b ; $\deg_P(v_i) = 2$ already, and a, b each already carry one path edge, so each admits at most one further edge. Brute force: $\max s(P)$ given $d(P) = 2$ is exactly 2 for $r = 2, 3$ and 1 for $r = 1$.
14. Max path weight $(2r-2)d(P) + s(P) \leq 4r-2$	VALID	$d=0$: $\leq 2r \leq 4r-2$ for $r \geq 1$; $d=1$: $\leq (2r-2) + 2r = 4r-2$; $d=2$: $\leq 2(2r-2) + 2 = 4r-2$. $\max(2r, 4r-2) = 4r-2$ for $r \geq 1$, checked symbolically. Brute force attains exactly $4r-2$ for $(h,r) = (5,1), (5,2), (7,1), (7,2), (7,3), (9,1), (9,2), (11,1)$.
15. Total weight $(2r-2)h + hr = (3r-2)h$	VALID	Arithmetic; D and S disjoint, all other edges weight 0 (and are explicitly allowed in paths, so no covering path is excluded).
16. Bound (5): $\pi(A) \geq \text{ceil}((3r-2)h/(4r-2))$	VALID	Standard weighted (LP-dual) counting: summing the weight of the covering paths is at least the total weight.

step	label	note
17. $(3r-2)/(4r-2) = \sqrt{3/4} - \sqrt{1/(8r-4)}$	VALID	Verified symbolically with sympy.
18. With $r = \sqrt{n} + 0(1)$, $h = n-r$ odd: $\pi(A) \geq 3n/4 - O(\sqrt{n})$	VALID	$(3/4 - \sqrt{1/(8r-4)})(n-r) = \sqrt{3n/4} - \sqrt{(n-r)/(8r-4)}$, and both correction terms are $O(\sqrt{n})$ at $r \approx \sqrt{n}$. Numerically, optimising r (the optimum is $r \approx \sqrt{n/6}$) gives $3n/4 - 0.61 \sqrt{n}$ for n up to 10^6 . Immediate from step 18. Quantified numerically: $c = 0.6$ from $n \geq 13$, $c = 0.7$ from $n \geq 120$, $c = 0.74$ from $n \geq 3600$, $c = 0.749$ from $n \geq 373753$.
19. “Every fixed $c < 3/4$ works for all sufficiently large n ”	VALID	Matches what is proven. No overclaim: the writeup does not assert that the constant $3/4$ is optimal, and does not claim to settle the paper’s Problem 1 (the $O(n)$ plane-matchings question), which Problem 9 was only a “first step” towards.
20. Verdict line: “an odd regular polygon with a central cluster yields ... at least $3n/4 - O(\sqrt{n})$ ”	VALID	

No step is labelled GAP or ERROR.

B.3 Reference check

The writeup **invokes no external result**: every geometric and combinatorial fact it uses is proved inline from the explicit coordinates. There is therefore no citation that could be misquoted, and `references_ok` is vacuously true. What I checked instead is the *ambient* literature (step 2b/2c of the referee instructions):

- **Source problem.** arXiv:2507.10840v2 fetched and text-extracted. Problem 9, Section 5, p. 8, matches the prompt verbatim (quoted above). The v2 comment (“Extended set of authors and strengthened results”) and journal reference Comb. Number Th. 15 (2026) 73–82 both confirmed.
- **Priority: no prior resolution found.** The Semantic Scholar citations API for arXiv:2507.10840 returns an empty list; OpenAlex has no record. Searches for the citing literature and for the construction itself turn up only the paper, its mirrors (people.maths.ox.ac.uk/~scott/Papers/monopaths.pdf, msp.org) and AI-summary aggregators.
- **State of the art on the weaker problem.** The best published lower bound for *partitioning* $\kappa_n[A]$ into plane **subgraphs** is the additive $n/2 + 1$ (Obenaus-Orthaber, arXiv:2108.05159, bumpy wheels; restated in Dumitrescu-Toth, arXiv:2405.17172, p. 8). No multiplicative improvement over $n/2$ is in the literature for either problem. Also checked and ruled out: Aichholzer et al. (arXiv:2112.08456), arXiv:2508.17277, arXiv:2510.22365, Pach-Saghafian-Schnider (arXiv:2306.13201), arXiv:2402.11044, arXiv:2507.06477 (“An Improved Bound for Plane Covering Paths”, a different problem).
- **A correction to the catalog blurb, not to the writeup.** The catalog page says “Pach-Saghafian-Schnider proved that every complete convex geometric graph decomposes into $n/2 + o(n)$ plane subgraphs”. That is a misreading of the source paragraph: the $n/2 + o(n)$ is the *authors’ own speculation* about plane subgraphs for general point sets, while PSS actually prove a *lower* bound of $n-1$ for decomposing the complete *convex* geometric graph into plane *star-forests*. The writeup does not repeat this error (it cites nothing), so this affects only the catalog entry.
- **Self-contamination check (2c).** The one web hit for the phrase “odd regular polygon with a central cluster ... $3n/4 - O(\sqrt{n})$ ” is github.com/graph-theory-AI/Graph-Theory-LLM-Proofs/pull/3, a PR opened 2026-09-16 by this very campaign, which already lists this claim keyed to 2507.10840__01. **That is this repository’s own output and is not prior art.** It was opened and confirmed to be AI-generated campaign material with no independent authors, venue or arXiv id. Excluding it, no independent source contains the construction or the bound. Priority is therefore clean, and the result is recorded as **not** ALREADY_KNOWN.

B.4 Computational check

Scripts: `verification_astra/scripts/2507.10840__01/`.

- `verify.py` (from the earlier, rate-limited run) - sympy exact algebraic arithmetic on the *true* regular polygon, with `selftest.py` confirming the predicate harness detects a planted degeneracy (even $h = 6$: v_0, o, v_3 collinear) and distinguishes endpoint contact from a proper crossing.
- `verify_rat.py` (**written for this review, independent implementation**) - the polygon vertices are *rational* approximations of the regular h -gon (denominator 10^7) and the cluster points are rational points in the disc of radius $\sin(\pi/(2h))/2$ about the origin. Every predicate is then an exact `Fraction` determinant, with no algebraic-number sign heuristic anywhere. Since Properties 1 and 2 are *open* conditions, a rational configuration that passes the tests is itself a legitimate witness - so the construction also works with rational coordinates. It uses its own branch-and-bound exact set-cover solver rather than `scipy’s milp`.
- `solver_selftest.py` - solver validation against a known value.
- `subgraph_consistency.py, asymptotics.py` - consistency and arithmetic checks.

Exact predicates (no three collinear; polygon convex in cyclic order; Property 1 in both directions; each cluster point strictly inside every T_i ; xz properly crossing $v_i a$ or $v_i b$ for *all* triples x, i, z): all pass for $(h, r) = (5, 1), (5, 2), (7, 1), (7, 2), (7, 3), (9, 1), (9, 2), (9, 3), (11, 1), (13, 1), (15, 3), (21, 4), (25, 5)$. Zero collinear triples in every case.

Exhaustive plane-path enumeration (all crossing-free simple paths, crossing tested exactly on every pair of edges):

h	r	n	max d(P)	max s(P) by d(P)	max abs weight	claim 4r-2	max abs $\setminus E(P) \cap (DUS) \setminus $
5	1	6	2	{0:2, 1:2, 2:1}	2	2	3
7	1	8	2	{0:2, 1:2, 2:1}	2	2	3
9	1	10	2	{0:2, 1:2, 2:1}	2	2	3
11	1	12	2	{0:2, 1:2, 2:1}	2	2	3
5	2	7	2	{0:4, 1:4, 2:2}	6	6	5
7	2	9	2	{0:4, 1:4, 2:2}	6	6	5
9	2	11	2	{0:4, 1:4, 2:2}	6	6	5
7	3	10	2	{0:6, 1:6, 2:2}	10	10	7

Every claim of the writeup holds **with equality**: $d(P) \leq 2$; $s(P) \leq 2r$; $d(P) = 2 \Rightarrow s(P) \leq 2$; max weight exactly $4r-2$; and $|E(P) \cap (DUS)| \leq 3$ for $r = 1$. The bounds are tight, not merely valid, so there is no slack hiding an error.

Exact $\pi(A)$ by exhaustive integer set cover over all maximal crossing-free paths. Two independent solvers were used: `scipy milp` on the sympy-exact regular-polygon geometry, and my own branch-and-bound on the rational geometry. For $n = 6$ and $n = 8$ both solvers were run and agree; $n = 10$ is the `scipy milp` result and $n = 7$ is my solver's. The independent branch-and-bound cross-check at $n = 10$ was attempted and hit its wall-clock cap without finishing, so that row rests on one solver rather than two (noted by the session after the referee was cut off mid-cleanup by a rate limit):

point set	n	maximal plane paths	exact $\pi(A)$	trivial bound $\text{ceil}(n/2)$	writeup's $\text{ceil}(2h/3)$
5-gon + centre	6	130	4	3	4
7-gon + centre	8	1449	5	4	5
9-gon + centre	10	15723	6	5	6
5-gon + 2 cluster pts	7	633	4	4	-

The construction **provably beats** $n/2$ already at $n = 6, 8, 10$, and the writeup's Section 2 bound $\text{ceil}(2h/3)$ is *exactly* the true value in all three cases. At $n = 6$ this is $\pi(A) = 4 = 2n/3$, matching Theorem part 1 at $q = 1$ on the nose.

Solver validation. On points in convex position my solver returns $\pi = 2, 3, 3, 4, 4$ for $n = 4, 5, 6, 7, 8$, i.e. exactly $\text{ceil}(n/2)$ - the known Bernhart-Kainen value. So the solver reproduces the known tight case and is not simply over-reporting.

Consistency with the literature record (the "too good to be true?" test). A cover by k plane *paths* yields a partition into $\leq k$ plane *subgraphs* (assign each edge to one path containing it), so $\chi_{\text{cross}}(A) \leq \pi(A)$ and never the reverse; a large π implies nothing about the subgraph problem. Computed exactly: for the 5-gon + centre, $\chi_{\text{cross}} = 3 = n/2$ while $\pi = 4$; for the 7-gon + centre, $\chi_{\text{cross}} = 4 = n/2$ while $\pi = 5$. So these point sets do **not** beat $n/2$ for plane subgraphs - no conflict with the $n/2 + 1$ record of Obenaus-Orthaber - and the gain is genuinely specific to the path problem. This removes the main reason to suspect the result was "too easy".

Asymptotics (sympy + exhaustive optimisation over r with $h = n-r$ odd): the identity $(3r-2)/(4r-2) = \sqrt{3/4} - \sqrt{1/(8r-4)}$ is exact; the optimal $r \approx \sqrt{n/6}$; the best bound is $3n/4 - 0.61 \sqrt{n}$ at $n = 10^6$ (0.60 - $0.61 \sqrt{n}$ throughout $n = 10^{2..10^6}$); $\text{ceil}(2(6q-1)/3) = 4q$ for all $q = 1..2000$.

B.5 Caveats

1. **Lower bound only.** The writeup determines no matching upper bound, so the true value of $\max_A \pi(A)$ is left open. Its own caveat line says as much. The $3/4$ is an artefact of the weighting ($d \leq 2, s \leq 2r, d = 2 \Rightarrow s \leq 2$), not a proven optimum.
2. **Wording nit in the general-position argument (step 11).** "avoiding the finitely many lines determined by previously chosen points" should say "determined by V and the previously chosen points". The conclusion is correct and I verified explicit rational witnesses exist up to $(h, r) = (25, 5)$, so nothing needs repairing in substance.
3. **"Sufficiently large n " is unquantified** in part 2 of the Theorem. This is harmless - part 1 already answers Problem 9 with $c = 2/3$ for every $n = 6q, q \geq 1$, and I verified $n = 6$ exactly - but for the record: $c = 0.7$ needs $n \geq 120$ and $c = 0.74$ needs $n \geq 3600$.
4. **Bound (5) requires $r \geq 2$** , as the writeup states. At $r = 1$ the D -weight $2r-2$ collapses to 0 and (5) degenerates to $\text{ceil}(h/2)$, weaker than Section 2's $\text{ceil}(2h/3)$; the two sections are separate arguments and Section 2 is not a special case of Section 3. The writeup is correct on this point but a reader could be misled by the layout.
5. U **shrinks with h** (it contains a disc of radius only $\sin(\pi/(2h)) \approx \pi/(2h)$). The writeup never needs a quantitative bound - U is a fixed nonempty open set for each fixed h , and r points in general position can always be placed in it - so the classic "sufficiently close with no quantified bound" failure mode does not bite here. I nevertheless instantiated it concretely (radius $\sin(\pi/(2h))/2$, rational coordinates) up to $h = 25$ to be sure.
6. **Degeneracy dependence on h odd is real and load-bearing**, not decorative: for even h the centre is collinear with antipodal vertex pairs, and the harness's self-test confirms the $h = 6$ configuration *is* degenerate. The writeup uses oddness correctly and in two places (general position, and the fact that the two D -edges at v_i are distinct).
7. **Scope.** Problem 9 is described in the source as "perhaps the first step towards" the paper's Problem 1 (an $O(n)$ upper bound on plane matchings/paths). Answering Problem 9 affirmatively does not resolve Problem 1 - they point in opposite directions. The writeup does not claim otherwise.
8. **Catalog metadata**, not the writeup: the problem is Problem 9 (Section 5), not "Problem 5.1", and the catalog's attribution of an $n/2 + o(n)$ convex upper bound to Pach-Saghafian-Schnider is incorrect (see Reference check).
9. **Novelty is verified only up to the limits of web search.** The result is not in any independent source I could find, and the sole apparent "prior art" is this campaign's own public PR, which I discounted per step 2c.

B.6 Referee summary

I set out to break this writeup and could not. The construction is an odd regular h -gon together with r points in the open region U common to all the triangles $\text{conv}\{v_i, v_{i+m}, v_{i+m+1}\}$. Its engine is a genuine simultaneous incompatibility: the h long chords $v_i v_{i+m}$

pairwise cross unless adjacent, so a plane path holds at most two of them and then only as a “wedge” $v_i v_{i+m}, v_i v_{i+m+1}$; but that wedge encloses the whole cluster, so every spoke from the cluster to any other polygon vertex would have to cross one of the two chords. A path can max out on chords or on spokes, never both. Weighting the chords by $2r-2$ and the spokes by 1 turns this into $\pi(A) \geq \text{ceil}((3r-2)h/(4r-2))$, i.e. $2n/3$ for $r = 1$, $n = 6q$ and $3n/4 - O(\sqrt{n})$ for $r \asymp \sqrt{n}$. All twenty numbered steps are VALID; the only blemish is a wording slip in the general-position argument that repairs itself. Every geometric predicate was re-checked in exact arithmetic - once on the true regular polygon with sympy algebraic numbers, and once, independently, on rational near-regular coordinates where every test is an integer determinant - for h up to 25 and r up to 5, and all of the writeup’s inequalities turn out to hold *with equality*, so no slack is concealing an error. Decisively, exhaustive integer set cover over all maximal crossing-free paths gives $\pi(A) = 4, 5, 6$ for $n = 6, 8, 10$ against the trivial bounds 3, 4, 5, reproducing $\text{ceil}(2h/3)$ exactly; the same solver returns the known $\text{ceil}(n/2)$ on convex point sets, so it is calibrated. The apparent “this is too easy” worry dissolves once one notes that $\pi(A)$ strictly dominates the plane-subgraph partition number - which for these very sets is only $n/2$ - so there is no conflict with the $n/2 + 1$ literature record. The problem is Problem 9 of arXiv:2507.10840 (the catalog’s “Problem 5.1”), quoted verbatim in the prompt; nothing cites that paper yet, no independent source contains this bound, and the single web hit for the construction is this campaign’s own GitHub pull request, which I discounted as self-contamination. **CONFIRMED**, high confidence: Problem 9 is answered affirmatively, with any $c < 3/4$.

References

- [1] A. Dumitrescu, J. Pach, M. Saghaian and A. Scott, Covering complete geometric graphs by monotone paths, Comb. Number Th. 15 (2026) 73–82; [arXiv:2507.10840](#) (v1 July 2025, v2 January 2026), Problem 9.
- [2] F. Bernhart and P. C. Kainen, The book thickness of a graph, J. Combin. Theory Ser. B 27 (1979) 320–331.
- [3] J. Obenaus and J. Orthaber, Edge partitions of complete geometric graphs (part 1), [arXiv:2108.05159](#) (2021).
- [4] O. Aichholzer, J. Obenaus, J. Orthaber, R. Paul, P. Schnider, R. Steiner, T. Taubner and B. Vogtenhuber, Edge partitions of complete geometric graphs (part 2), [arXiv:2112.08456](#) (2021).
- [5] J. Pach, M. Saghaian and P. Schnider, Decomposition of geometric graphs into star forests, [arXiv:2306.13201](#) (2023).
- [6] A. Dumitrescu and J. Pach, Partitioning complete geometric graphs on dense point sets into plane subgraphs, [arXiv:2405.17172](#) (2024); Graph Drawing and Network Visualization (GD 2024), LIPIcs 320, paper 9.
- [7] H. A. Akitaya, G. Aloupis, A. Biniaz, P. Bose, J.-L. De Carufel, C. Gavoille, J. Iacono, L. Kleist, M. Smid, D. Souvaine and L. Theocharous, An improved bound for plane covering paths, [arXiv:2507.06477](#) (2025), ESA 2025.